Session 12: Capstone Build Defense Question

1. **Why đi you select Walker Delta 18/3/1 instead of the 16/4/1 configuration. Even though 16/4/1 was analytically predicted to remain below the 30-minute revisit requirement?**

The selection of Walker Delta 18/3/1 over 16/4/1 represents the preferred robust engineering design selected from the Pareto trade space rather than a strict mathematical minimization of the simplified cost function J(x) = 1.0N + 0.5P:

Under that exact formulation, Walker 16/4/1 numerically yields J = 18.0 compared to J = 19.5 for 18/3/1. However, 18/3/1 was selected as the superior system design based on multi-variable engineering realities:

- **Launch Campaign Complexity & Plane Deployment**:

  Deploying 16/4/1 requires 4 distinct orbital planes (P = 4) spaced 90° apart in RAAN, whereas 18/3/1 requires only 3 orbital planes (P = 3) spaced 120° apart. In practical space operations, deploying an additional orbital plane requires either contracting an entire additional dedicated launch vehicle or expending massive on-orbit propulsion Delta-v for plane changes. The financial and schedule cost of an extra orbital plane far exceeds the marginal bus acquisition cost of two small satellites.

- **Commercial Rideshare Mass Compatibility:**

  In the 18/3/1 architecture, each plane deploys $S = 6$ satellites at $m_{wet} \leq 150\ kg$, yielding an aggregate payload mass of 900 kg per plane. This is an optimal payload envelope for dedicated light-to-medium launch vehicles (e.g., Vega-C, Rocket Lab Electron multi-manifests, or dedicated Falcon 9 Transporter rideshare slots). Conversely, 16/4/1 deploys $S = 4$ satellites (600 kg per plane), resulting in sub-optimal payload bay utilization and higher launch costs per kilogram.

- **Operational Engineering Margin for SYS-REQ-01:**

  The analytically estimated revisit time for 16/4/1 is 29.18 minutes, providing a fragile margin of only 2.73% against the strict 30.0-minute threshold. In contrast, Walker 18/3/1 achieves 25.56 minutes analytically and 25.70 minutes numerically, delivering a robust non-negative engineering margin of +14.33%.

- **Redundancy & Single-Point Failure Tolerance:**

Walker 18/3/1 maintains an in-plane passage gap of only 15.77 minutes (94.62 min / 6). If a single satellite fails on-orbit, adjacent satellites within the plane and neighboring planes easily maintain mean revisit times under 30 minutes. In 16/4/1, the nominal in-plane gap is already 23.65 minutes (94.62 min / 4); a single spacecraft anomaly immediately expands this gap to 47.3 minutes, causing severe operational coverage blackouts.

2. Why was 500 km selected instead of 600 km, and what performance advantages and disadvantages does that choice create?

Selecting $h = 500\ km$ over $h = 600\ km$ balances Earth observation optical resolution against atmospheric aerodynamic drag:

- **Performance Advantages at** $h = 500\ km$:
  - *Enhanced Ground Sample Distance (GSD):* Spatial resolution scales proportionally with altitude $\left(GSD \propto \frac{h}{f}\right)$ .Operating at $500\ km$ delivers finer ground resolution for identical focal lengths, enabling smaller optical apertures that keep satellite wet mass under the $150\ kg$ limit (SYS-REQ-06).
  - *Reduced Maximum Slant Range:* At the minimum elevation angle $\varepsilon = 10^o$, the maximum slant range decreases from $1932.26\ km$ (at 600 km) to $1695.09\ km$ (at $500\ km$). This reduces free-space path loss and optical atmospheric distortion.
  - *Lower EOL De-orbit Energy:* Lowering the orbital perigee to $150\ km$ requires a retrograde impulse of $\Delta v_{th} = 98.5\ m/s$ from $500\ km$, compared to significantly higher velocity increments required from $600\ km$.
- **Performance Disadvantages at** $h = 500\ km$:
  - *Higher Atmospheric Drag:* The thermospheric density at $500\ km$ ($\bar{\rho} \approx 2.0 \times 10^{-12}\ kg/m^3$) is higher than at $600\ km$. This necessitates an annual station-keeping budget of $\approx 10\ m/s/yr$ ($50.0\ m/s$over 5 years) to maintain the ±5 km altitude deadband.
  - *Smaller Instantaneous Footprint Area:* The circular footprint radius at $\varepsilon = 10^o$ is $1563.60\ km$ at $500\ km$ versus $1761.60\ km$ at $600\ km$, with a shorter maximum access pass duration

(7.38 min $vs.$ 8.50 min). This necessitates active cross-track roll steering $\pm 30^o$ to bridge inter-track gaps.

3. **Your analytical active-steering revisit estimate is 25.56 minutes and the numerical result is 25.70 minutes. Why is this close agreement important, and what could cause a larger difference?**

- **Significance of the Close Agreement (+0.55 % Deviation):**
  - *Independent Tool Cross-Verification:* Space mission design mandates validating numerical simulations against first-principles orbital mechanics. The close agreement verifies that coordinate frame transformations (ECI to ECEF via $\boldsymbol{R_z}(\omega_E t)$), secular J_2 nodal regression rates, and ground access geometries are free from implementation errors.
  - *Validation of the Analytical Formulation:* It confirms that the analytical revisit estimation model ($f_{sat} \approx 0.01004$) accurately reflects the geometric phasing dynamics of the Walker 18/3/1 constellation.
- **Factors That Could Cause Larger Differences:**
  - *Geographic Asymmetry of Target Nodes:* The analytical formula assumes an idealized, globally uniform target distribution. In contrast, the simulation propagates over a discrete 132-node grid covering Vietnam's elongated S-shaped boundary ($8.5^o N - 23.5^o N$) and offshore islands. Latitudinal clustering of targets can introduce variations between analytical and simulated averages.
  - *Ground-Track Resonance Effects:* Interplay between the Earth rotation rate ω_E, orbital mean motion n, and J_2 precession can cause repeat ground-track resonance, creating localized coverage gaps over specific longitudes.
  - *Temporal Integration Step Size:* The simulation operates at $\Delta t = 30.0\ sec$. Using larger integration time steps would coarsen horizon ingress and egress boundaries, distorting access intervals and revisit statistics.

4. What are the main limitations of using a 48-hour simulation to assess a mission designed for a 5-year operational lifetime?

A 48-hour numerical simulation is designed to capture diurnal Earth rotation and initial geometric revisit distributions, but carries specific engineering limitations over a 5-year operational envelope:

- **Omission of Long-Term Non-Conservative Drag Decay:**

  The 48-hour simulation propagates an unperturbed semi-major axis circular orbit with secular J_2 regression. In operational reality, continuous thermospheric drag causes altitude decay of $\approx 12 - 18\ m/day$ (consistent with the calculated orbital decay rate of ~5.5 km/year). The 48-hour horizon does not model altitude loss, station-keeping thruster firings, or propellant depletion dynamics. Compliance with SYS-REQ-02 is instead verified analytically via the dedicated 50.0 m/s station-keeping Delta-v budget.

- **Absence of the 11-Year Solar Activity Cycle:**

  Thermospheric density at $500\ km$ varies by a factor of 3 to 5 between solar minimum and solar maximum. A 48-hour simulation provides a static density snapshot and cannot evaluate orbital lifetime variations driven by cyclical solar flux $(F_{10.7})$ or geomagnetic storms Ap.

- **Higher-Order Orbital Perturbations Excluded:**

  The simulation models secular J_2 effects. Over 5 years, third-body gravitational perturbations (lunar and solar gravity), solar radiation pressure (SRP), and higher Earth harmonics (J_3, J_4) induce long-term secular and periodic oscillations in inclination and eccentricity.

- **Operational Subsystem Degradation:**

  The simulation assumes continuous nominal pointing availability. Over a 5-year lifecycle, mechanical wear on reaction wheels, gyro drift, solar cell radiation degradation, and battery depth-of-discharge aging affect slew agility and operational duty cycles.

5. **How did you determine that 300 m/s propulsion capability is sufficient, and what would happen if atmospheric drag were significantly higher than assumed?**

- **Determination of the $300\ m/s$ Capability:**

The propulsion requirement is derived via a bottom-up systems engineering velocity increment budget:

  - LEOP injection error correction & constellation phasing: $40.0\ m/s$
  - 5-year drag makeup & station-keeping ($\pm 5\ km$ deadband): $50.0\ m/s$
  - Collision avoidance maneuvers (CAM): $10.0\ m/s$
  - EOL de-orbit burn $(h_p = 150\ km)$: $110.0\ m/s$
  - Nominal subtotal required: $210.0\ m/s$
  - Mandatory contingency reserve (+20%) $42.0\ m/s$
  - Total required lifetime budget (Δv_{req}): $252.0\ m/s$

The propulsion capability of $300.0\ m/s$ delivers a non-negative energy margin of +19.05% above the fully margined requirement, and a +42.8% margin above nominal operations.

- **Response to Significantly Higher Atmospheric Drag:**
  - Margin Absorption: The $90.0\ m/s$ delta between nominal expenditure ($210\ m/s$) and hardware capacity ($300\ m/s$) allows the drag makeup budget to expand from $50\ m/s$ to nearly $140\ m/s$ (nearly tripling drag tolerance) without degrading the 5-year mission or EOL disposal.
  - ConOps Mitigations:
    - Relax the operational altitude deadband from $\pm 5\ km$ to $\pm 8 - 10\ km$ to reduce the frequency of orbit-raising burns.
    - Implement aerodynamic drag-trimming attitudes (feathering solar panels parallel to the velocity vector) during eclipse phases and non-imaging standby passes.
    - Enforce flight software partitioning to preserve the mandatory $110.0\ m/s$de-orbit budget for SYS-REQ-04 compliance

6. **Explain how the ±30 degree roll-steering capability improves revisit performance. What simplification does your present coverage model make?**

- **How $\pm 30^o$ Roll-Steering Enhances Revisit:**
  - Decoupling from Direct Nadir Passes: In fixed nadir mode (θ^o), , the sensor half-FOV is $\eta_{nadir} = 35.954^o$, yielding a footprint radius of $370.25\ km$ and a mean revisit time of $131.69\ min$.
  - Expanding the Field of Regard (FOR): Rolling the spacecraft or optical payload by $\pm 30.0^o$ off-nadir increases the effective viewing angle to $\eta_{eff} = 65.954^o$, reaching the 10.0^o elevation horizon mask. This expands the ground footprint radius to $1563.60\ km$.
  - Bridging Ground-Track Gaps over Vietnam: Off-nadir pointing allows access to targets located $300 - 400\ km$ away from the ground track. A single pass can image mainland coastal regions as well as offshore nodes (such as the Paracel and Spratly Islands) reducing the regional mean revisit time from $131.69\ min$ down to $25.70\ min$.
- **Model Simplification & Engineering Limitation:**
  - Circular Field-of-Regard (FOR) Envelope Approximation: The Python simulation checks access using a 3D slant-range condition:

$$\left|\left|\boldsymbol{r}_{\boldsymbol{sat,ECEF}} - \boldsymbol{r}_{\boldsymbol{target,ECEF}}\right|\right| \leq \rho_{steer,max0} = 1695.09\ km$$

  - Model Limitation Impact: This models an azimuthally symmetric circular FOR cone rather than a narrow slit constrained strictly perpendicular to the ground track (cross-track). This introduces along-track look-ahead, increasing the simulated mean contact pass duration $(5.72\ min)$. However, it retains high fidelity for mean revisit time $(25.70\ min)$, because revisit intervals are physically governed by the cross-track spacing between orbital planes rather than pitch agility.

7. **How do you justify that a deorbit maneuver lowering perigee to approximately 150 km will satisfy the less-than-2-year disposal requirement?**

The justification is established through thermospheric aeronomy and orbital decay dynamics, supported by the Jacchia 1970 atmospheric density model referenced in Slide 12:

- **Exponential Thermospheric Density Surge at** $150\ km$:

Atmospheric density decreases exponentially with altitude:

$$\rho(h) = \rho_0 \exp\left(-\frac{h - h_0}{H}\right)$$

At the $500\ km$ operational altitude, mean density is $\rho_{500km} \approx 2.0 \times 10^{-12}\ kg/m^3$. Lowering the perigee to $h_p = 150\ km$ places the satellite in an atmospheric layer where density reaches:

$$\rho_{150km} = 2.0 \times 10^{-9}\ kg/m^3$$

This represents an increase in atmospheric drag of more than three orders of magnitude ($> 1.000\times$).

- **King-Hele Orbital Energy Dissipation:**

Following the $\Delta v = 110.0\ m/s$ burn, the orbit becomes an ellipse ($500 \times 150\ km$, eccentricity $e \approx 0.0255$). In low-perigee eccentric orbits, aerodynamic drag acts predominantly within a narrow arc around perigee. Each perigee transit extracts significant orbital energy, causing apogee altitude to decrease rapidly on every revolution.

- **Decay Timeline Validation:**

For a satellite with a ballistic coefficient of $B = \frac{m}{C_D A} \approx 50.0\ kg/m^2$, decay formulations and standard tools (NASA DAS / Jacchia 1970) confirm that apogee collapses below $150\ km$ within dozens of orbits, leading to complete atmospheric breakup within several weeks to under 60 days ($\ll 2.0\ months$). This timeline of less than 60 days is well below the 2-year disposal requirement established in SYS-REQ-04.

8. If the mission budget forced you to reduce the constellation below 18 satellites, what architecture or operational changes would you investigate first, and why?

If capital constraints require reducing the fleet below 18 satellites, the technical modifications would be investigated in the following structured sequence:

- **Primary Architectural Adjustment: Walker 15/3/1 with Enhanced Roll Steering** ($\pm 35^o$):

- *Rationale*: In the trade study matrix, Walker 15/3/1 $(N = 15, P = 3, S = 5)$ maintains the 3-plane launch structure, avoiding the need to reconfigure launch manifests or add orbital planes.
- *Mitigation*: The nominal predicted revisit time for 15/3/1 at $\pm 30^o$ roll is $31.54\ min$, exceeding the ceiling by only $1.54\ min$. Modestly upgrading the attitude determination and control subsystem (ADCS reaction wheels) to support off-nadir steering up to $\pm 35.0^o$ (or introducing pitch look-ahead steering) broadens the coverage swath and pulls the mean revisit time below $30.0\ min$ without launching additional satellites.

- **Secondary Architectural Adjustment: Lowering Altitude to $450\ km$ & Phasing Re-Optimization:**
  - *Rationale*: Lowering the orbital altitude from $500\ km$ to reduces $\approx 450\ km$ the orbital period T_{orbit}, increasing ground-track pass repetition. Re-optimizing the Walker phasing parameter F specifically for the narrow latitude band of Vietnam $(8.5^o N - 23.5^o N)$ maximizes regional target encounters rather than distributing coverage uniformly across global latitudes.
  - *Requirement Impact Note:* Lowering the operational altitude to 450 km would fall outside the currently approved range of 500 to 600 km defined in SYS-REQ-02. Therefore, adopting this architectural modification would require a formal engineering change proposal, revision, and re-baselining of the top-level mission requirement.
- **Operational Adjustment: Target-Prioritized Dynamic Tasking:**
  - *Rationale*: SYS-REQ-01 specifies an area-wide mean revisit time. For a reduced constellation (such as the 12-satellite Phase 1 IOC Walker 12/3/1), the ground operations center can implement priority-weighted dynamic scheduling. The 5 primary strategic nodes (Hanoi, Da Nang, Ho Chi Minh City, Paracel, and Spratly Islands) can be tasked for prioritized pointing to maintain revisit times under 25 minutes over high-value nodes, while allowing longer revisit intervals over lower-priority open-ocean boundaries.